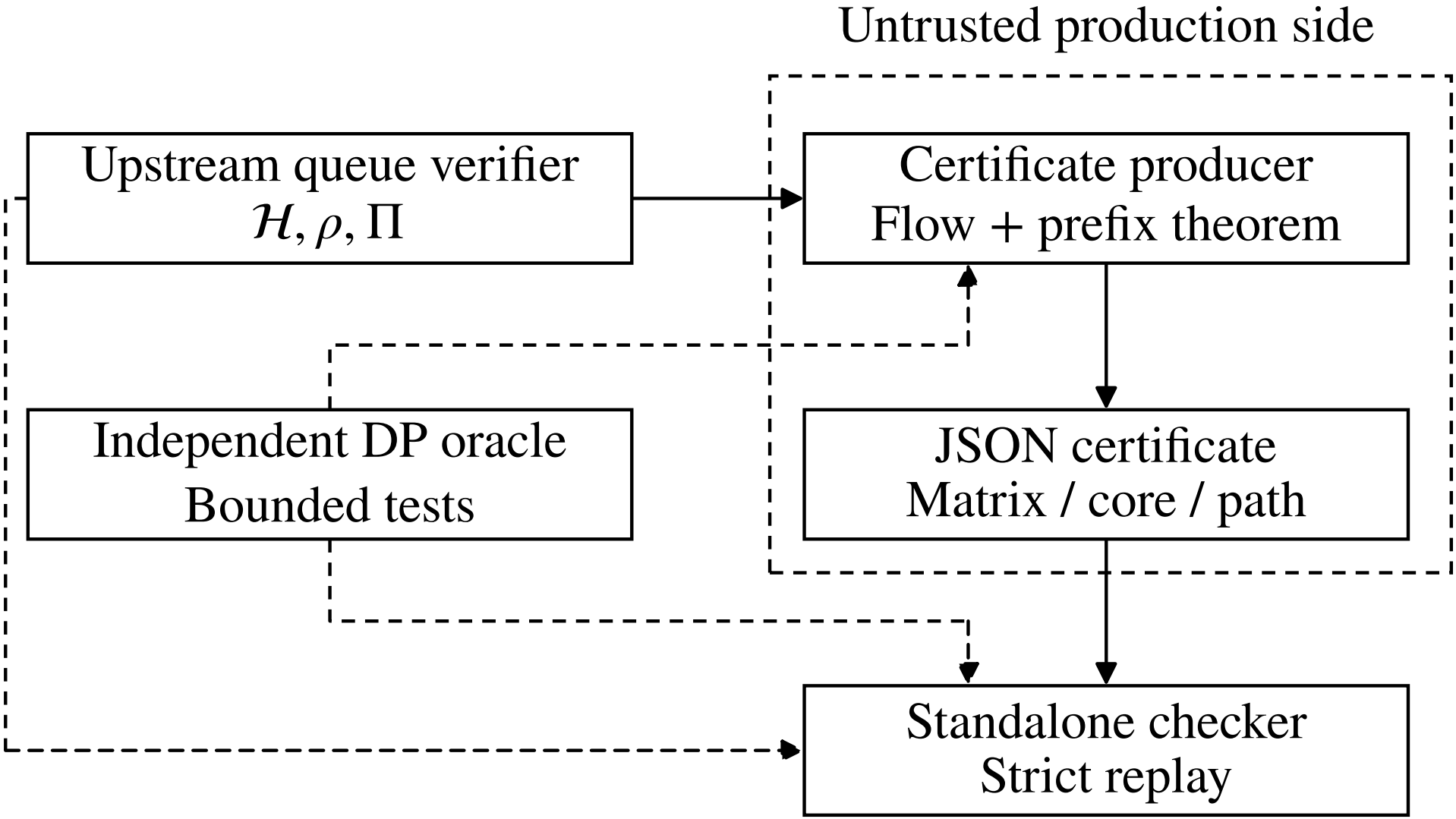
Untrusted production side
Upstream queue verifier
$\mathcal{H}, \rho, \Pi$
Certificate producer
Flow + prefix theorem
Independent DP oracle
Bounded tests
JSON certificate
Matrix / core / path
Standalone checker
Strict replay